Reflection in C++26 (P2996)

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What is reflection?

> *"The ability of software to expose its internal structure."*

Static reflection: the **compiler** exposes structure at compile time

- Inspection of types, members, functions, and annotations
- Automatic C++ code generation at compile time
- Expressive compile-time metaprogramming without macros

P2996 — one of the **largest** proposals in C++ history since the introduction of templates.

Why should you care?

One C++ description → *everything*:

- **Automatic serialization / deserialization**
- **Generic struct printing / debugging**
- **Compile-time generation of types via splicers**
- **Doc generation** (md, html…)
- Annotations (P3394, paired with reflection) let you attach compile-time metadata to declarations and read it back via reflection, supporting things like validation rules or serialization hints.
- …

The two operators in one picture

```
constexpr std::meta::info info = ^^Circle;             // reflect-on a type
typename [: info :] c = {.name = "c", .radius = 1.0}; // splice it back into code
// also works on members, expressions, namespaces, templates...
```

`^^` aka - cat-ears operator

- Everything that crosses the boundary is `consteval` / `constexpr`
- `^^T` lifts the *name* `T` into a constexpr value (`std::meta::info`)
- `[: info :]` drops a `std::meta::info` value back into a *type*, *expression*, or *member*

Why not earlier?

Why not earlier? — C++ libraries pre-P2996

Library	Type	Intrusive?	Coverage
RTTI (`typeid`)	language	none	type identity only
Boost.PFR	header	none	aggregates only
magic_enum	header	none	enums (via `__PRETTY_FUNCTION__`)
RTTR	header	macro	fields, methods, inheritance
EnTT::meta	header	registration	data + funcs (games)
Boost.Describe	header	macro	members + bases
Qt MOC	codegen	`Q_OBJECT`	full + signals/slots
Unreal UHT	codegen	`UCLASS`	full + GC + Blueprint
SWIG	codegen	none	multi-language bindings

Why not earlier? — every other language has this

Language	Runtime	Compile-time	Idiom
Python	🟢 deep	⚪	`inspect`, `getattr`, decorators
Java	🟢 native	🟡 processors	`java.lang.reflect`
JavaScript	🟢 Proxy	⚪	`Reflect`, `Object.*`
Rust	🔴 minimal	🟢 proc-macros	`derive`, `serde`, `bevy_reflect`
C++26	🟡 build manually	🟢 P2996	`^^T`, `[: r :]`, `consteval`

C++ was **the only major language without first-class reflection** — until now.

Supported compilers

Compiler	Reflexion C++26	`template for` (expansion statement)
GCC 16.1+	✅ Almost done	✅
Clang (Bloomberg)	✅ Almost done	✅ (`-freflection-latest`)
Clang mainline	🟡 Partial	🟡 In progress
MSVC	❌	❌
EDG	🟡 Partial	🟡

A running example

```cpp
struct Point {
    int    x;
    int    y;
    double z;
    double norm() const { return ...; }
    Point scale(double) const { return ...; }
};
```

Enumerate `Point`'s fields

```cpp
constexpr auto ctx = std::meta::access_context::current();

constexpr auto fields = std::define_static_array(std::meta::nonstatic_data_members_of(^^Point, ctx));

template for (constexpr auto f : fields) { // expansion statement
    std::println(
        "  {} : {}",
        std::meta::identifier_of(f),
        std::meta::display_string_of(std::meta::type_of(f))
    );
}
```

Output:

```
  x : int
  y : int
  z : double
```

No macros, no inheritance, no registration, non intrusif. Just the type.

Enumerate `Point`'s methods

```
constexpr auto members = std::define_static_array(std::meta::members_of(^^Point, ctx));

template for (constexpr auto m : members) { // expansion statement
    if constexpr (is_exportable_member_function(m)) {
        std::print("  {} {}(", std::meta::display_string_of(std::meta::return_type_of(m)),
                   std::meta::identifier_of(m));

        bool first = true;
        template for (constexpr auto param: std::define_static_array(std::meta::parameters_of(m))) {
            if (!first) {std::print(", ");}
            first = false;
            std::print("{}", std::meta::display_string_of(std::meta::type_of(param)));
        }
        std::println(")");
    }
}
```

Output:

```
  double norm()
  Point scale(double)
```

Simple JSON-ish serialization

```cpp
template <typename T>
std::string to_json(const T& obj) {
    std::string out = "{";
    bool first = true;
    template for (constexpr auto m : std::define_static_array(
                      std::meta::nonstatic_data_members_of(^^T))) {
        if (!first) out += ",";
        first = false;
        out += '"';
        out += std::meta::identifier_of(m);
        out += "\":";
        out += serialize_value(obj.[:m:]);  // user per-type value formatter
    }
    out += "}";
    return out;
}
```

Serialization example

```
enum class Color { Red, Green, Blue };

struct Address {
    std::string street;
    int         number;
};

struct Person {
    std::string              name;
    int                      age;
    bool                     active;
    Color                    favorite_color;
    Address                  home;
    std::vector<std::string> tags;
};

int main() {
    Person p{
        .name           = "Alice",
        .age            = 30,
        .active         = true,
        .favorite_color = Color::Green,
        .home           = {"Rue Pasteur", 42},
        .tags           = {"admin", "ops"},
    };

    std::println("{}", to_json(p));
}
```

Serialization result


```json
{
    "name": "Alice",
    "age": 30,
    "active": true,
    "favorite_color": "Green",
    "home": {
        "street": "Rue Pasteur",
        "number": 42
    },
    "tags": [
        "admin",
        "ops"
    ]
}
```

Aggregate example

Use `std::meta::define_aggregate`:
(previsouly called `std::meta::define_class` in draft)

```cpp
#include <meta>

struct Synthesized;  // incomplete — to be defined

consteval {
    std::meta::define_aggregate(^^Synthesized, {
        std::meta::data_member_spec(^^int,    {.name = "id"}),
        std::meta::data_member_spec(^^double, {.name = "value"}),
    });
}

// Synthesized is now:
struct Synthesized {int id; double value;};
```

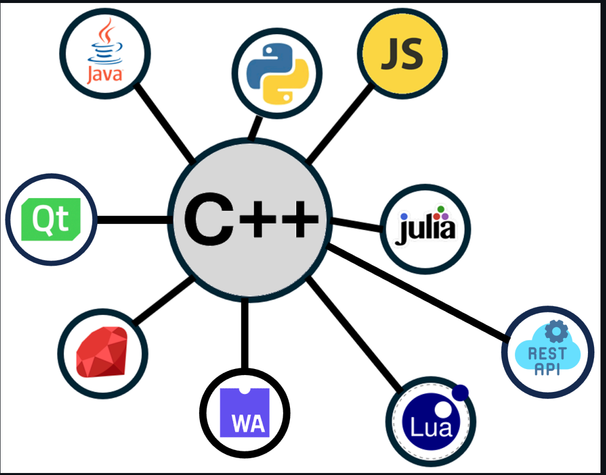
Etonish, nein ? 😃 (cf, "*La Minute nécessaire de monsieur Cyclopède*")

Better Etonish!

```
constexpr auto v = [: parse_json(json_data) :];
```

Means: "Run `parse_json(json_data)` at **compile time**, obtain a reflected representation of some C++ entity or expression, and substitute the corresponding C++ code here."

The example of Rosetta



- A C++ introspection & automatic language binding.
- Non intrusif
- Introspection **at runtime**
- Hand written registration à la `pybind11`

https://github.com/xaliphostes/rosetta2

Rosetta — *before* C++26

For each class, hand-write the registration:

```
rosetta::Class<Shape>("Shape")
    .field("name", &Shape::name).doc("display name")
    .method("describe", &Shape::describe).doc("greeting")
    .method("area",     &Shape::area)
    .static_method("next_id", &Shape::next_id);

rosetta::Class<Circle>("Circle")
    .base<Shape>()
    .field("radius", &Circle::radius)
        .doc("radius in meters")
        .range(0.0, 1e6);

// ... repeat for Rectangle, every other type, every project ...
```

Real numbers from our internal lib: **~6000 lines** of glue.

Rosetta — *with* C++26

Use C++ annotations (P3394) for doc, readonly, numeric range, alias, displayed-name...

```
rosetta::register_reflected<Shape>();
rosetta::register_reflected<Circle>();
rosetta::register_reflected<Rectangle>();
```

That's it:

- Same registry
- Simplified downstream generators

~100 lines of `register_reflected` machinery — once, in the library — and *you never touch it again*!

Rosetta before and after C++26 (# lines of code)

Spec	Before	After
core	5,903	106
common	2,380	0
js	1,808	204
py	1,553	90
rest	3,327	305
wasm	1,435	102
Total	16,500	807

Applications — Auto language bindings

One C++ description, **N hosts** (visitors):

Target	What reflection drives
Python	`pybind11` `def(...)`, `__init__`, dunder methods
JavaScript	Node.js N-API wrappers, getters/setters
WebAssembly	Emscripten `EMSCRIPTEN_BINDINGS`
TypeScript	`.d.ts` declaration files
REST API	HTTP routes mapped to methods, JSON I/O
Lua / Ruby / Julia	Sol2, Rice, CxxWrap
Java/JNI, C#/.NET, Swift	bridge generation

The alternative is **N hand-maintained binding files** that drift as the C++ API evolves.

Applications — Validation · Docs · Persistence · Live

- **Validation** — range, regex, not-null, cross-field invariants (from annotations).
- **Documentation** — Markdown / HTML / OpenAPI / Sphinx from `doc("…")`.
- **Persistence / ORM (object relational mapping)** — `CREATE TABLE`, migrations from class diffs.
- **Configuration & DI (dependency injection)** — config-file → object hydration, CLI flag generation.
- **Live scripting / REPL (read-eval-print loop)** — every method auto-callable from the console.
- **Testing** — property-based, fuzzing, snapshot tests, binding-coverage.

Applications — Serialization · GUI · REST

Serialize: JSON / XML / YAML / TOML / Protobuf / … — *no schema needed.*

GUI generation:

- Qt property editors (`ObjectInspector<Circle>` for free)
- QML data binding, web forms, Blueprint-style node editors
- ImGui debug panels — sliders, color pickers, drag-floats per field
- …

REST / RPC: each method → endpoint, each parameter → JSON field.

```
for (auto cls : registry.list_classes())
    for (auto m : cls->methods())
        router.add(cls->name + "/" + m.name, &dispatch);
```

Annotation is the leverage

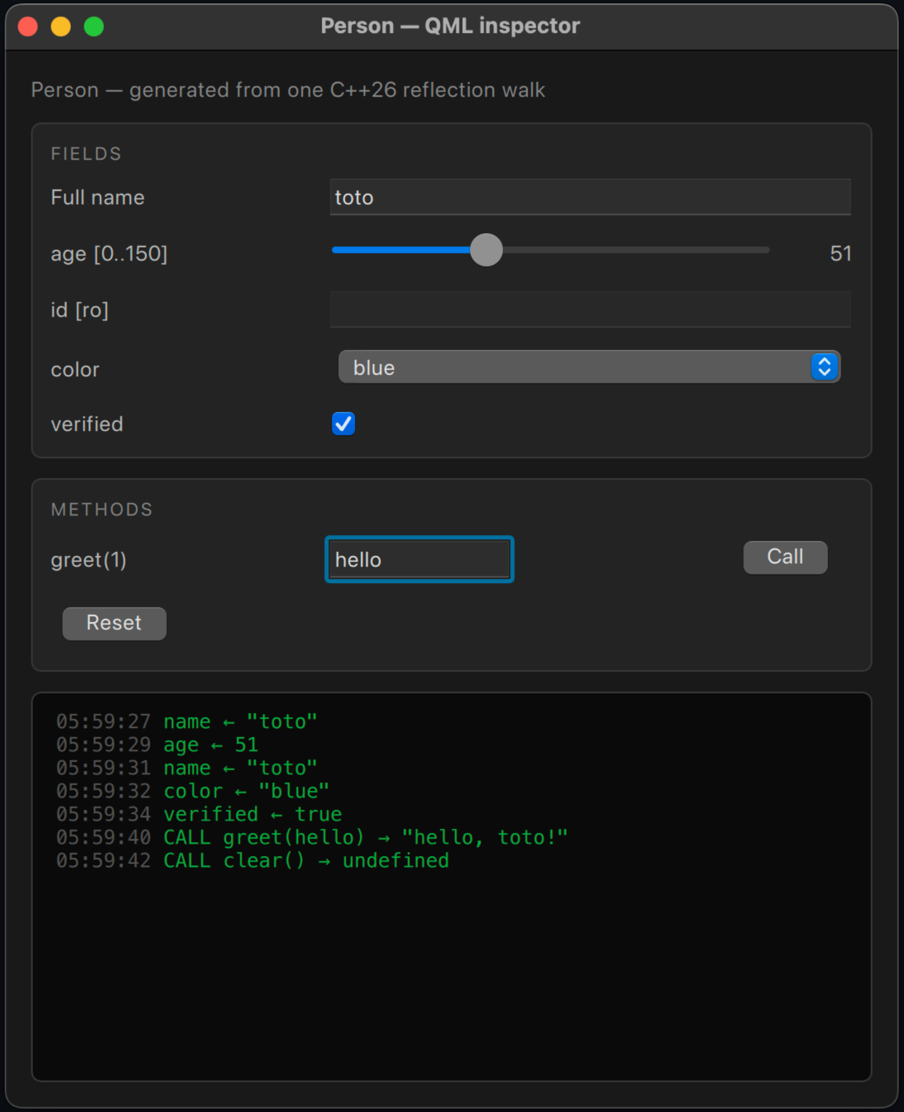
One `[[=doc{...}, =range{...}]] double radius;` is **simultaneously**:

- A Python attribute
- A JavaScript getter/setter
- A REST endpoint
- A JSON-serializable element
- A Qt property in the inspector
- A documented entry in the reference
- A range-validated database column
- ...

> **Every line of registration unlocks features across multiple subsystems at once.**

QML example

```cpp
using namespace rosetta;

struct Person {
    [[ = doc{"the person's display name"} ]]
    std::string name;

    [[ = doc{"age in whole years"},
       = range{0.0, 150.0},
       = widget::slider ]]
    int age = 0;

    [[ = doc{"server-assigned identifier"},
       = readonly{} ]]
    std::string id;

    [[ = rosetta::doc{"favourite colour"},
       = rosetta::combobox{{"red", "green", "blue", "yellow"}} ]]
    std::string color = "red";

    Person() = default;
    Person(std::string n, int a, std::string i): name(std::move(n)), age(a), id(std::move(i)) {}

    [[ = doc{"Returns a greeting prefixed by the given salutation."} ]]
    std::string greet(const std::string &salutation) const {
        return salutation + ", " + name + "!";
    }

    void clear() {
        name.clear();
        age = 0;
    }
};

Person person;
rosetta::ReflectedObject reflected;
rosetta::bind_qml<Person>(&reflected, person);

QQmlApplicationEngine engine;
engine.rootContext()->setContextProperty("inspector", &reflected);
```



Qt example

```
class Algo {
public:

    [[ = rosetta::doc{"Tolerence of the solver"},
       = rosetta::range{1e-10, 1e-6},
       = rosetta::widget::textfield, = rosetta::label{"EPS"} ]]
    double eps{1e-7};

    [[ = rosetta::doc{"Set the max iterations"},
       = rosetta::range{0, 200},
       = rosetta::widget::slider, = rosetta::label{"Max iter"} ]]
    int maxIter{100};

    [[ = rosetta::doc{"Tell if the solver must be iterative"},
       = rosetta::widget::checkbox,
       = rosetta::label{"Iterative solver"} ]]
    bool iterative{true};

    [[ = rosetta::doc{"Solver name"},
       = rosetta::combobox{{"Seidel", "Jacobi", "gmres", "cgnr"}},
       = rosetta::label{"Solver name"} ]]
    std::string solverName{"Seidel"};

    [[ = rosetta::doc{"Run the solver and return the convergence"},
       = rosetta::button{"Run"} ]]
    double run() {return 1e-8;}

    [[ = rosetta::doc{"Reset the solver"},
       = rosetta::button{"Reset"} ]]
    void reset() {}
};

Algo algo;
QWidget *inspector = rosetta::build_inspector<Algo>(algo, "Algo");
```

Doc generation example

```cpp
#include <rosetta/docgen.h>
#include "Algo.h"

const auto md = rosetta::generate_markdown<Algo>();
```

Algo

Fields

Name	Type	Description
`eps`	`double`	Tolerence of the solver (range: 1e-10..1e-06)
`maxIter`	`int`	Set the max iterations (range: 0..200)
`iterative`	`bool`	Tell if the solver must be iterative
`solverName`	`std::string`	Solver name (choices: Seidel, Jacobi, gmres, cgnr)

Methods

`run() → double`

Run the solver and return the convergence

`reset() → void`

Playing with C++26 to recreate the Qt `moc`

Qt's moc is a separate code generator. It parses headers, emits .moc files, links them in. It's been a Qt fixture for 25 years.

C++26 reflection makes it... optional.

```
using namespace rosetta;

class Thermostat {
public:
    [[ = moc::signal ]]
    moc::Signal<double> temperatureChanged;

    [[ = moc::property{"temperature", "temperatureChanged"} ]]
    double m_temperature = 20.0;
};
```

No .moc file. No build step. Just a header.

```cpp
Thermostat th;
Display    d;

moc::connect<"temperatureChanged", "showTemperature">(th, d);

moc::set<"temperature">(th, 19.8); // fires temperatureChanged(19.8)
moc::set<"temperature">(th, 19.8); // equality-gated, silent
```

Three failure modes, all at compile time:

- typo'd signal name → static_assert: "no [[=signal]]-tagged member"
- typo'd slot name → static_assert: "no [[=slot]]-tagged member"
- mismatched types → template error at the lambda body

References

- Reflection proposal
- Python Bindings with Value-Based Reflection
- Using Reflection to Replace a Metalanguage for Generating JS Bindings
- Short video presentation for bindings

Questions?

Example Python binding

```
template <typename T> void bind(py::module_ &m, const char *py_name) {
    py::class_<T> cls(m, py_name);
    cls.def(py::init<>());

    // Fields
    template for (constexpr auto fld : std::define_static_array(std::meta::nonstatic_data_members_of(^^T, ctx))) {
        if constexpr (ro) {
            // readonly annotation -> Python read-only property
            cls.def_property_readonly(name, [](const T &self) -> MemberT {
                return self.[:fld:]; }, docstr);
            }
    }
    // Methods
    template for (constexpr auto fn : std::define_static_array(std::meta::members_of(^^T, ctx))) {
        if constexpr (is_exportable_member_function(fn)) {
            constexpr auto name  = std::define_static_string(std::meta::identifier_of(fn));
            constexpr auto m_doc = std::meta::annotation_of_type<doc>(fn);
            constexpr const char *mdoc = m_doc.has_value() ? m_doc->text : "";

            if constexpr (std::meta::is_static_member(fn)) {
                cls.def_static(name, &[:fn:], mdoc);
            } else {
                cls.def(name, &[:fn:], mdoc);
            }
        }
    }
}
```